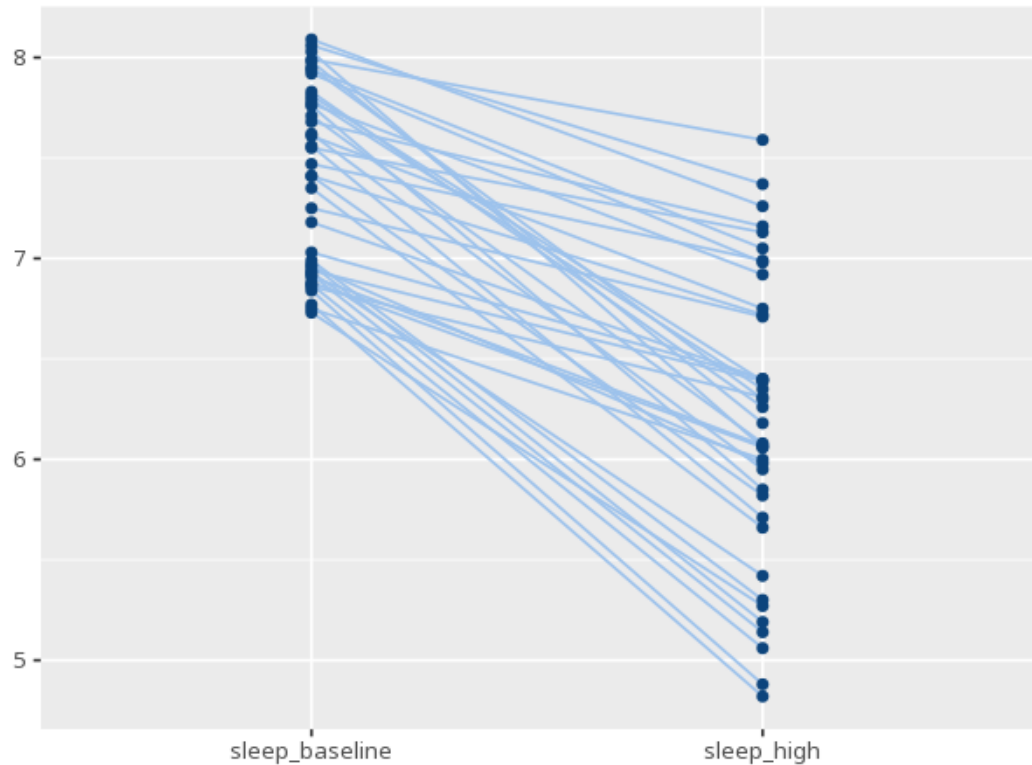


1 Paired t-test

Condition	N	M	SD
sleep_baseline	40	7.40	0.45
sleep_high	40	6.20	0.72

Pair	t	df	p	Mdiff	95% CI	d [95% CI] _z	r (paired)
sleep_baseline – sleep_high	14.14	39	<.001	1.20	[1.03, 1.38]	2.24 [1.65, 2.81]	0.66



Tests whether the average change between two related measurements (e.g. before vs. after) differs from zero. A p below alpha means a significant change; the mean difference, CI and d_z describe its size (their sign depends on the subtraction order, $A-B$). A p at or above alpha does not prove there was no change — only that none was detected. Your significance threshold (α) is 0.05.

A paired-samples t-test gave $M=1.2$, $t(39)=14.14$, $p < .001$, $d_z=2.24$ [1.65, 2.81].

Statistical basis: Paired-samples t-test (Student (1908). "The probable error of a mean." *Biometrika*, 6(1), 1-25.); Cohen's d_z effect size (Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Routledge.); Paired correlation (Pearson r) between the two conditions (Pearson, K. (1895). "Note on regression and inheritance in the case of two parents." *Proceedings of*

the Royal Society of London, 58, 240-242.).